

Manuscript Add-On Sections for the Updated Learning Workflow

NN_FPN repository notes

June 14, 2026

1 Updated learning workflow for the filtered P_N model

The original filtered P_N solver is retained, but the learned filter is now trained and selected by a sweep-driven workflow. For a moment order N , let $u_i^n = (u_{i,0}^n, \dots, u_{i,N}^n)^\top$ denote the cell-centered moment vector and let $\phi_i^n = u_{i,0}^n$ denote the scalar flux. The neural filter is a cellwise nonnegative map

$$\sigma_{f,i}^n = F_{\theta,N}(x_i^n) \geq 0,$$

where x_i^n is the feature vector defined below. The constant-filter baselines use a single learned scalar $c_N \geq 0$, i.e.

$$\sigma_{f,i}^n = c_N.$$

The learned filter strength multiplies the same filter coefficients used in the filtered P_N update, so the discretization and evaluation metric remain those of the manuscript.

1.1 Training objective

For a training draw ω , let $\phi_{N,\theta}^{FPN}(x_i, T_\omega; \omega)$ be the filtered P_N scalar flux produced by the learned filter and let $\phi^{\text{ex}}(x_i, T_\omega; \omega)$ be the high-order reference solution. The updated neural-network and constant-filter sweep models minimize the relative final-time scalar-flux loss

$$\mathcal{L}_N(\theta) = \frac{1}{B} \sum_{b=1}^B \frac{\frac{1}{n_x} \sum_{i=1}^{n_x} \left(\phi_{N,\theta}^{FPN}(x_i, T_b; \omega_b) - \phi^{\text{ex}}(x_i, T_b; \omega_b) \right)^2}{\max\left(\frac{1}{n_x} \sum_{i=1}^{n_x} (\phi^{\text{ex}}(x_i, T_b; \omega_b))^2, \varepsilon\right)}, \quad \varepsilon = 10^{-12}.$$

No auxiliary moment loss is used in the reported updated runs. The scalar metric reported in the result tables is unchanged from the manuscript,

$$r_N = \frac{\left\| \phi_N^{FPN} - \phi^{\text{ex}} \right\|_2}{\left\| \phi_N^{P_N} - \phi^{\text{ex}} \right\|_2},$$

so smaller values indicate stronger error reduction relative to the unfiltered P_N solution.

1.2 Training distributions

The paper-setting baseline uses the original four-family training distribution, which combines Gaussian, Heaviside, bump, and discontinuous-source initial/source families with random homogeneous material coefficients. The updated sweep-selected workflow replaces this with an augmented distribution $\mathcal{D}_{\text{aug}}$. A draw $\omega \sim \mathcal{D}_{\text{aug}}$ samples the initial condition, source, and material profile from a mixture that includes Gaussian, discontinuous, vanishing-cross-section-like, source-driven, and Reeds-like cases. This changes the training ensemble only; all reported test cases and the definition of r_N are unchanged.

2 Input feature map replacing the baseline network inputs

For each cell i , define the discrete transport residual and material/source terms

$$d_i = \frac{(Au)_{i+1/2} - (Au)_{i-1/2}}{\Delta x}, \quad c_i = \sigma_{t,i} u_i, \quad s_i = \sigma_{s,i} u_{i,0}, \quad q_i = Q_i,$$

where $d_i, c_i \in \mathbb{R}^{N+1}$ and $s_i, q_i \in \mathbb{R}$. For any cellwise channel tensor $g_{i,k}$, let

$$\mathcal{N}_x(g)_{i,k} = \frac{g_{i,k} - \frac{1}{n_x} \sum_{j=1}^{n_x} g_{j,k}}{\text{std}_{1 \leq j \leq n_x}(g_{j,k}) + 10^{-10}}$$

be the per-sample spatial normalization. The baseline manuscript-style feature vector is

$$z_i^{\text{base}} = [\mathcal{N}_x(|d_i|), \mathcal{N}_x(|c_i|), \mathcal{N}_x(|s_i|), \mathcal{N}_x(|q_i|)] \in \mathbb{R}^{2N+4}.$$

The sweep also introduced log-magnitude material and residual features,

$$L_{\alpha,[a,b]}(g) = \text{clip}_{[a,b]} \left(\log \left(1 + \frac{|g|}{\alpha} \right) \right), \quad \sigma_{a,i} = \max(\sigma_{t,i} - \sigma_{s,i}, 0),$$

and the material vectors

$$m_i^{\log} = [L_{\alpha,[a,b]}(\sigma_{s,i}), L_{\alpha,[a,b]}(\sigma_{t,i}), L_{\alpha,[a,b]}(\sigma_{a,i})],$$

$$m_i^{\text{all}} = \left[m_i^{\log}, \frac{\sigma_{s,i}}{\max(|\sigma_{t,i}|, \varepsilon)}, \frac{\sigma_{a,i}}{\max(|\sigma_{t,i}|, \varepsilon)} \right].$$

The selected feature maps are

$$\begin{aligned} x_i^{(3)} &= [z_i^{\text{base}}, m_i^{\text{all}}], & \alpha &= 0.1, \quad [a, b] = [0, 20], \quad \dim x_i^{(3)} = 15, \\ x_i^{(7)} &= [L_{0.1,[0,10]}(d_i), L_{0.1,[0,10]}(c_i), L_{0.1,[0,10]}(s_i), L_{0.1,[0,10]}(q_i), \mathcal{N}_x(m_i^{\text{all}})], & [a, b] &= [0, 10], \quad \dim x_i^{(7)} = 23, \\ x_i^{(9)} &= [z_i^{\text{base}}, \mathcal{N}_x(m_i^{\log})], & \alpha &= 1.0, \quad [a, b] = [0, 40], \quad \dim x_i^{(9)} = 25. \end{aligned}$$

The $N = 3$ and $N = 9$ maps retain the normalized baseline residual/source features, while the $N = 7$ map uses clipped log-magnitude residual/source features without residual/source normalization.

3 Selected training settings

The hyperparameters below are the settings selected by the sweep and then retrained for 10 independent replicas. Each run uses Adam and the final-time scalar-flux loss above.

N	hidden	batch	epochs	lr	scheduler	wd NN	wd const	feature map	horizons
3	64	32	100	0.003	constant	10^{-3}	10^{-4}	$x_i^{(3)}$	{0.5}
7	512	64	100	0.01	cosine, warmup 0.05	10^{-6}	10^{-4}	$x_i^{(7)}$	{0.5}
9	256	32	100	0.01	constant	10^{-6}	10^{-4}	$x_i^{(9)}$	{0.25, 0.5, 1.0}

For comparison, the paper-setting constant-filter baseline uses the original paper training distribution, 10 replicas, batch size 64, 200 epochs, Adam with learning rate 0.1, cosine scheduling with warmup fraction 0.05, and weight decay 10^{-6} for each $N \in \{3, 7, 9\}$.

4 Updated one-dimensional results

Tables report mean $\pm$ standard deviation over 10 independently trained replicas. Here r_N^{NN} denotes the updated neural network, $r_N^{\text{const, aug}}$ denotes a constant filter trained on the same augmented sweep-selected setting as the neural network, and $r_N^{\text{const, paper}}$ denotes the constant filter trained with the original paper setting. Bold entries are the lowest mean value among the three learned-filter variants at fixed N , test case, and time.

4.1 Gaussian

t_f	$N = 3$			$N = 7$			$N = 9$		
	r_3^{NN}	$r_3^{\text{const, aug}}$	$r_3^{\text{const, paper}}$	r_7^{NN}	$r_7^{\text{const, aug}}$	$r_7^{\text{const, paper}}$	r_9^{NN}	$r_9^{\text{const, aug}}$	$r_9^{\text{const, paper}}$
0.5	0.2467 $\pm$ 0.0098	0.9814 $\pm$ 0.0010	0.5772 $\pm$ 0.0009	0.1089 $\pm$ 0.0169	0.9618 $\pm$ 0.0012	0.5325 $\pm$ 0.0012	0.1108 $\pm$ 0.0788	0.9543 $\pm$ 0.0042	0.5344 $\pm$ 0.0013
1.0	0.2096 $\pm$ 0.0023	0.9647 $\pm$ 0.0018	0.3636 $\pm$ 0.0010	0.1036 $\pm$ 0.0400	0.9237 $\pm$ 0.0023	0.2790 $\pm$ 0.0013	0.0363 $\pm$ 0.0222	0.9101 $\pm$ 0.0081	0.2793 $\pm$ 0.0014

4.2 Vanishing cross section

t_f	$N = 3$			$N = 7$			$N = 9$		
	r_3^{NN}	$r_3^{\text{const, aug}}$	$r_3^{\text{const, paper}}$	r_7^{NN}	$r_7^{\text{const, aug}}$	$r_7^{\text{const, paper}}$	r_9^{NN}	$r_9^{\text{const, aug}}$	$r_9^{\text{const, paper}}$
0.5	0.3339 $\pm$ 0.0091	0.9835 $\pm$ 0.0009	0.6267 $\pm$ 0.0008	0.0881 $\pm$ 0.0200	0.9613 $\pm$ 0.0012	0.5252 $\pm$ 0.0012	0.1040 $\pm$ 0.0749	0.9552 $\pm$ 0.0041	0.5330 $\pm$ 0.0014
1.0	0.0817 $\pm$ 0.0048	0.9621 $\pm$ 0.0020	0.2975 $\pm$ 0.0012	0.0896 $\pm$ 0.0401	0.9263 $\pm$ 0.0022	0.2841 $\pm$ 0.0014	0.0274 $\pm$ 0.0181	0.9106 $\pm$ 0.0081	0.2734 $\pm$ 0.0015

4.3 Discontinuous cross section

t_f	$N = 3$			$N = 7$			$N = 9$		
	r_3^{NN}	$r_3^{\text{const, aug}}$	$r_3^{\text{const, paper}}$	r_7^{NN}	$r_7^{\text{const, aug}}$	$r_7^{\text{const, paper}}$	r_9^{NN}	$r_9^{\text{const, aug}}$	$r_9^{\text{const, paper}}$
0.5	0.3151 $\pm$ 0.0083	0.9829 $\pm$ 0.0009	0.6114 $\pm$ 0.0008	0.1050 $\pm$ 0.0208	0.9607 $\pm$ 0.0012	0.5234 $\pm$ 0.0012	0.1093 $\pm$ 0.0977	0.9547 $\pm$ 0.0042	0.5311 $\pm$ 0.0014
1.0	0.1991 $\pm$ 0.0088	0.9623 $\pm$ 0.0020	0.2942 $\pm$ 0.0012	0.1479 $\pm$ 0.0382	0.9242 $\pm$ 0.0023	0.2772 $\pm$ 0.0014	0.0480 $\pm$ 0.0227	0.9094 $\pm$ 0.0082	0.2703 $\pm$ 0.0015

4.4 Reeds

t_f	$N = 3$			$N = 7$			$N = 9$		
	r_3^{NN}	$r_3^{\text{const, aug}}$	$r_3^{\text{const, paper}}$	r_7^{NN}	$r_7^{\text{const, aug}}$	$r_7^{\text{const, paper}}$	r_9^{NN}	$r_9^{\text{const, aug}}$	$r_9^{\text{const, paper}}$
5.0	0.4493 $\pm$ 0.0099	0.8638 $\pm$ 0.0066	0.2613 $\pm$ 0.0002	0.4725 $\pm$ 0.0813	0.7413 $\pm$ 0.0067	0.2502 $\pm$ 0.0002	0.3582 $\pm$ 0.0330	0.7037 $\pm$ 0.0225	0.2438 $\pm$ 0.0002
10.0	0.7526 $\pm$ 0.0128	0.8215 $\pm$ 0.0077	0.5550 $\pm$ 0.0000	0.8206 $\pm$ 0.1134	0.5277 $\pm$ 0.0093	0.4007 $\pm$ 0.0006	0.5745 $\pm$ 0.0728	0.4807 $\pm$ 0.0296	0.3242 $\pm$ 0.0005